PRIME DENSITY OF ISOMORPHIC REDUCTION GROUPS FOR PRIME-ISOGENOUS ELLIPTIC CURVES WITH NON-CM FINITE-LEVEL EVALUATION

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

1.1. **Preliminaries.** Let E and E' be isogenous elliptic curves defined over a finite field k. In a series of papers [3, 4, 5, 6] we considered variants of the following question.

Question 1. *If $E(k) \simeq E'(k)$, is it true that $E(K) \simeq E'(K)$ as K varies over finite extensions of k?*

The answer is no, and it is not hard to come up with explicit counterexamples. However, in [3] the first-named author showed that if ℓ is an odd prime number and E and E' are ℓ-isogenous, where the kernel is generated by a k-rational point of order ℓ, then $E(k) \simeq E'(k)$ implies that $E(K) \simeq E'(K)$ for all finite extensions K of k. If $\ell = 2$ then it was shown in [4] if $E(k) \simeq E'(k)$ *and* $E(\mathbf{k}) \simeq E'(\mathbf{k})$, where $\mathbf{k}$ is the unique quadratic extension of k, then $E(K) \simeq E'(K)$ for all finite extensions K of k.

Building on this, in [5] and [6] we fixed 2-isogenous elliptic curves E and E' over $\mathbf{Q}$ and defined a prime p to be *anomalous* for the pair (E, E') if $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$, but $E(\mathbf{F}_{p^2}) \not\simeq E'(\mathbf{F}_{p^2})$. The main result of these papers is that the set of anomalous primes has a well-defined density that can be realized by summing a certain geometric series associated to the 2-adic representations of E and E'. For example, if the 2-adic representations of E and E' are as large as possible (since they are 2-isogenous, each has a rational 2-torsion point, so the images have index 3 in $\mathrm{GL}_2(\mathbf{Z}_2)$), then the anomalous primes for (E, E') have density $1/30$.

In the course of determining the density of anomalous primes for all pairs (E, E') of 2-isogenous elliptic curves, there is a simpler question that we did not answer:

Question 2. *Let ℓ be a prime number. If E and E' are ℓ-isogenous elliptic curves, what is the proportion $P(E, E')$ of primes p for which $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$?*

More precisely, we consider the ratio

$$P(E, E')[X] = \frac{\{p \leq X \ : \ E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)\}}{\pi(X)},$$

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where π is the prime-counting function, and its limit $\lim_{X\to\infty} P(E,E')[X]$. When this limit exists, we denote it by $P(E,E')$. It is a consequence of our prior and current work that the limit always exists and can be computed using the Chebotarev Density Theorem.

1.2. **Main Theorem.** Let E be an elliptic curve over $\mathbf{Q}$. Fix an algebraic closure $\overline{\mathbf{Q}}$ of $\mathbf{Q}$ and write $\mathrm{Gal}_{\mathbf{Q}}$ for the absolute Galois group. If ℓ is a prime number, then we write $T_\ell E$ for the ℓ-adic Tate module of E and

$$\rho_{E,\ell} : \mathrm{Gal}_{\mathbf{Q}} \to \mathrm{Aut}(T_\ell E), \text{ and}$$
$$\overline{\rho}_{E,\ell^n} : \mathrm{Gal}_{\mathbf{Q}} \to \mathrm{Aut}(T_\ell E \otimes \mathbf{Z}/\ell^n\mathbf{Z})$$

for the ℓ-adic and mod ℓ^n representations of E, respectively. If $G \subseteq \mathrm{GL}_2(\mathbf{Z}_\ell)$ is the image of the ℓ-adic representation, then we write $G(\ell^n) \subseteq \mathrm{GL}_2(\mathbf{Z}/\ell^n\mathbf{Z})$ for its reduction modulo ℓ^n. If E' is ℓ-isogenous to E then we write G' and $G'(\ell^n)$ for the images of the ℓ-adic and mod ℓ^n representations of E', respectively. Since E and E' are ℓ-isogenous over $\mathbf{Q}$, it follows that $G(\ell)$ and $G'(\ell)$ are Borel subgroups of $\mathrm{GL}_2(\mathbf{Z}/\ell\mathbf{Z})$. The level of the ℓ-adic representation is ℓ^M where M is the smallest positive integer for which G is the full preimage of $G(\ell^M)$ in $\mathrm{GL}_2(\mathbf{Z}_\ell)$.

Let $L_{\ell^m} = \mathbf{Q}(E[\ell^m])$ and $L'_{\ell^m} = \mathbf{Q}(E'[\ell^m])$ be the ℓ^m-division fields of E and E', respectively. Then L_{ℓ^m} and L'_{ℓ^m} are Galois over $\mathbf{Q}$ with Galois groups $G(\ell^m)$ and $G'(\ell^m)$, respectively. If p is a prime number, let F and F' denote matrix representatives of the Frobenius classes of E and E', respectively, as elements of $\mathrm{GL}_2(\mathbf{Z}_\ell)$. Then F and F' will define elements in $G(\ell^m)$ and $G'(\ell^m)$ by reduction modulo ℓ^m.

Definition 1.1. *Let $m \geq 1$. We define $d_{\ell^m} \in [0,1]$ to be the proportion of prime numbers such that $F' \not\equiv I \pmod{\ell^m}$ given that $F \equiv I \pmod{\ell^m}$. Similarly, we define d'_{ℓ^m} to be the proportion of prime numbers such that $F \not\equiv I \pmod{\ell^m}$ given that $F' \equiv I \pmod{\ell^m}$.*

In Section 2 we determine the possible values of d_{ℓ^m} and d'_{ℓ^m} by interpreting conditional probabilities in terms of the degrees of the composite field extensions $[L_{\ell^m}L'_{\ell^m} : L_{\ell^m}]$ and $[L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}]$. Depending on the level of the ℓ-adic images of E and E', once m is sufficiently large it will follow from our work below that $d_{\ell^m} = d'_{\ell^m} = 1 - 1/\ell$. In order to calculate $P(E,E')$, it is then a matter of determining $[L_{\ell^m}L'_{\ell^m} : L_{\ell^m}]$, $[L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}]$, $|G(\ell^m)|$, and $|G'(\ell^m)|$ for small m and then summing a geometric series.

If E and E' do not have CM, then $[\mathrm{GL}_2(\mathbf{Z}_\ell) : G] = [\mathrm{GL}_2(\mathbf{Z}_\ell) : G']$ by [7, Prop. 2.1.1]. Therefore, if the maximum of the levels of their ℓ-adic representations is ℓ^M then $|G(\ell^M)| = |G'(\ell^M)|$. By the classification of non-CM ℓ-adic images in [16], once $\ell > 5$ the level of the ℓ-adic representations of E and E' is ℓ (see also [8]). If the level is ℓ then, as will follow from our work below, $P(E,E')$ is completely determined by $|G(\ell)|$ and $|G(\ell')|$ (which must be equal). If $\ell = 5$ then there are a handful of special cases where the level is 25; here one would need to do separate calculations to determine d_5 and d'_5, but then the calculation of $P(E,E')$ would proceed similarly to the level ℓ case. If $\ell = 2$ or 3, then there are many possibilities for the 2- and 3-adic Galois images, as well as for the values of d_{ℓ^m} and d'_{ℓ^m}, for small values of m (we can always take $m \leq 5$). However, these field extensions are small enough that one could, in principle, through routine computation, determine all possible $P(E,E')$ for all pairs of 2- and 3-isogenous elliptic curves over $\mathbf{Q}$ using

the classification of 2-adic images in [15] and 3-adic images in [16]. The general result on the values of the d_{ℓ^m} and d'_{ℓ^m} is as follows.

Proposition 1.2. *With notation as above, suppose the elliptic curves E and E' do not have CM and that the maximum of the levels of their ℓ-adic representations is ℓ^M. Then we have $d_{\ell^m}, d'_{\ell^m} \in \{0, 1-1/\ell\}$ for all $m \geq 1$ and $d_{\ell^m} = d'_{\ell^m} = 1-1/\ell$ for all $m \geq M$.*

Theorem 1.3. *Let E and E' be ℓ-isogenous elliptic curves over $\mathbf{Q}$. Then the proportion $P(E, E')$ of primes such that $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$ is given by the formula*

$$P(E,E') = 1 - \sum_{m=1}^{\infty} \left(\frac{d_{\ell^m}}{|G(\ell^m)|} + \frac{d'_{\ell^m}}{|G'(\ell^m)|} \right). \tag{1.4}$$

Remark 1.5. Theorem 1.3 holds for all pairs of ℓ-isogenous elliptic curves over $\mathbf{Q}$, including those with CM. However, there are some significant differences in the CM case that make the evaluation of $P(E, E')$ much more straightforward. We will show in Section 4 below that $P(E, E') = 1$ once $\ell > 3$. If $\ell = 2$ or 3, then we can have $P(E, E') \neq 1$ but, as we will also show, there are only a handful of possibilities for $P(E, E')$. This complements [6, Thm. 1.11] where we proved that if E and E' are 2-isogenous CM elliptic curves over $\mathbf{Q}$, then the proportion of anomalous primes is either $1/12$ or 0.

Because the techniques used in this paper are very similar to those of [5] and [6], we aim for brevity by referring the reader, whenever possible, to those background results rather than rederiving them here. Throughout the paper we use the LMFDB [12] for specific examples and Pari/GP [14] for computation.

2. Main Result

If E and E' are ℓ-isogenous elliptic curves defined over a finite field $\mathbf{F}_p$, then the prime-to-ℓ parts of the groups $E(\mathbf{F}_p)$ and $E'(\mathbf{F}_p)$ are isomorphic [3, Cor. 1]. Suppose the ℓ-Sylow subgroup of $E(\mathbf{F}_p)$ has size ℓ^v. If $E(\mathbf{F}_p)$ and $E'(\mathbf{F}_p)$ are nonisomorphic, then by the theory of isogeny volcanoes (see [17] for background) there exists a positive integer $m \leq v/2$ such that

$$E(\mathbf{F}_p)[\ell^\infty] = \mathbf{Z}/\ell^m\mathbf{Z} \times \mathbf{Z}/\ell^{v-m}\mathbf{Z}, \text{ and}$$
$$E'(\mathbf{F}_p)[\ell^\infty] = \mathbf{Z}/\ell^{m-1}\mathbf{Z} \times \mathbf{Z}/\ell^{v-m+1}\mathbf{Z},$$

or

$$E(\mathbf{F}_p)[\ell^\infty] = \mathbf{Z}/\ell^m\mathbf{Z} \times \mathbf{Z}/\ell^{v-m}\mathbf{Z}, \text{ and}$$
$$E'(\mathbf{F}_p)[\ell^\infty] = \mathbf{Z}/\ell^{m+1}\mathbf{Z} \times \mathbf{Z}/\ell^{v-m-1}\mathbf{Z}.$$

In the first case, we have $F \equiv I \pmod{\ell^m}$ and $F' \not\equiv I \pmod{\ell^m}$, while in the second case we have $F' \equiv I \pmod{\ell^{m+1}}$ and $F' \not\equiv I \pmod{\ell^{m+1}}$. For the rest of the paper we fix a positive integer $m \geq 1$. We now identify two proportions of prime numbers, stated in terms of the Frobenius classes to which they belong.

Definition 2.1. *Let $X, X' \in [0,1]$ be defined as the following conditional probabilities:*

$$X = \mathrm{Prop}\left(F' \equiv I \pmod{\ell^m} \mid F \equiv I \pmod{\ell^m}\right), \text{ and}$$
$$X' = \mathrm{Prop}\left(F \equiv I \pmod{\ell^m} \mid F' \equiv I \pmod{\ell^m}\right).$$

By the Chebotarev Density Theorem,

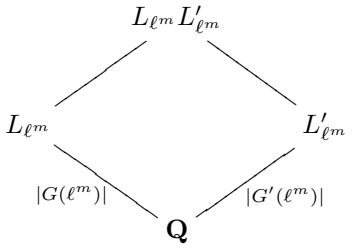
$$\begin{aligned}\text{Prop}\,(F \equiv F' \equiv I \pmod{\ell^m}) &= \frac{1}{[L_{\ell^m}L'_{\ell^m} : \mathbf{Q}]},\\ \text{Prop}\,(F \equiv I \pmod{\ell^m}) &= \frac{1}{[L_{\ell^m} : \mathbf{Q}]} = \frac{1}{|G(\ell^m)|}, \text{ and}\\ \text{Prop}\,(F' \equiv I \pmod{\ell^m}) &= \frac{1}{[L'_{\ell^m} : \mathbf{Q}]} = \frac{1}{|G'(\ell^m)|}.\end{aligned}$$

By basic field theory

$$[L_{\ell^m}L'_{\ell^m} : \mathbf{Q}] = [L_{\ell^m}L'_{\ell^m} : L_{\ell^m}] \cdot |G(\ell^m)| = [L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}] \cdot |G'(\ell^m)|.$$

Lemma 2.2. *With all notation as above, we have* $X = [L_{\ell^m}L'_{\ell^m} : L_{\ell^m}]^{-1}$ *and* $X' = [L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}]^{-1}$.

Proof. This is Bayes' Law applied to the Chebotarev Density Theorem and the field diagram



□

Since

$$[L_{\ell^m}L'_{\ell^m} : \mathbf{Q}] = [L_{\ell^m}L'_{\ell^m} : L_{\ell^m}] \cdot |G(\ell^m)| = [L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}] \cdot |G'(\ell^m)|,$$

it follows that

$$\frac{[L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}]}{[L_{\ell^m}L'_{\ell^m} : L_{\ell^m}]} = \frac{|G(\ell^m)|}{|G'(\ell^m)|} \in \{1, \ell, 1/\ell\}, \tag{2.3}$$

with $[L_{\ell^m}L'_{\ell^m} : L_{\ell^m}], [L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}] \in \{1, \ell\}$. For proofs of these statements, see [6, §2].

Lemma 2.4. *With all notation as above, we have* $d_{\ell^m} = 1 - X$ *and* $d'_{\ell^m} = 1 - X'$.

Proof. This follows directly from the definitions of $d_{\ell^m}, d'_{\ell^m}, X$, and X'. □

Proposition 2.5. *Suppose* E *and* E' *do not have CM and that the maximum of the levels of their* ℓ*-adic representations is* ℓ^M*. Then if* $m \geq M$*, we have*

$$[L_{\ell^m}L'_{\ell^m} : L_{\ell^m}] = [L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}] = \ell.$$

Proof. Since the images of the ℓ-adic representations of E and E' have level at most ℓ^M, G and G' are the full preimages of $G(\ell^M)$ and $G'(\ell^M)$ in $\mathrm{GL}_2(\mathbf{Z}_\ell)$, respectively. Fix $m \geq M$. We show that $[L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}] \neq 1$ by proving that there exists a prime p for which $E(\mathbf{F}_p)[\ell^m] \cong \mathbf{Z}/\ell^m\mathbf{Z} \times \mathbf{Z}/\ell^m\mathbf{Z}$ but $E'(\mathbf{F}_p)[\ell^m] \ncong \mathbf{Z}/\ell^m\mathbf{Z} \times \mathbf{Z}/\ell^m\mathbf{Z}$. We follow a strategy similar to the proof of [5, Theorem 5.2.4]. We prove that there exists a prime p for which the height of the ℓ-isogeny volcano containing E is exactly m, there is a unique vertex on the crater of the volcano, and this vertex corresponds

to E. In this case, the isogeny $E \to E'$ must be descending. Switching the roles of E and E' then shows that $[L_{\ell^m} L'_{\ell^m} : L_{\ell^m}] \neq 1$.

We consider odd ℓ and $\ell = 2$ separately. Suppose ℓ is an odd prime and let δ be a quadratic nonresidue modulo ℓ. Since ℓ^m is at least the ℓ-adic level of E, there exists a prime p such that $F \equiv \left(\begin{smallmatrix} 1 & \ell^m \\ \delta\ell^m & 1 \end{smallmatrix}\right) \pmod{\ell^{2m+1}}$. Since $t \equiv 2 \pmod{\ell^{2m+1}}$ and $p \equiv \det(F) \equiv 1 - \delta\ell^{2m} \pmod{\ell^{2m+1}}$, we see that

$$t^2 - 4p \equiv 4 - 4(1 - \delta\ell^{2m}) \equiv 4\delta\ell^{2m} \pmod{\ell^{2m+1}}.$$

This implies $v_\ell(t^2 - 4p) = 2m$.

Let $\mathcal{O}_0$ be the endomorphism ring of an elliptic curve lying on the crater of the ℓ-isogeny volcano corresponding to E, $D_0 = \operatorname{disc}(\mathcal{O}_0)$, and $\mathcal{O}_K$ be the maximal order of $\mathbf{Q}(\sqrt{t^2 - 4p})$. We now apply a result of Kohel to determine several important properties of this volcano [17, Thm. 7]. Since $\ell \nmid [\mathcal{O}_K : \mathcal{O}_0]$, the height of this volcano is exactly m. Since

$$\left(\frac{D_0}{\ell}\right) = \left(\frac{4\delta}{\ell}\right) = \left(\frac{\delta}{\ell}\right) = -1,$$

there is a unique vertex on the crater of the volcano. Since $F \equiv I \pmod{\ell^m}$, we see that the vertex corresponding to E must be this unique vertex on the crater of the volcano. This completes the proof in the case where ℓ is an odd prime.

Now suppose $\ell = 2$. Since 2^m is at least the 2-adic level of E, there exists a prime p such that $F \equiv \left(\begin{smallmatrix} 1+2^m & 2^m \\ 2^m & 1 \end{smallmatrix}\right) \pmod{2^{2m+3}}$. Since $t \equiv 2 + 2^m \pmod{\ell^{2m+3}}$ and $p \equiv \det(F) \equiv 1 + 2^m - 2^{2m} \pmod{2^{2m+3}}$, we see that

$$t^2 - 4p \equiv 4 + 4 \cdot 2^m + 2^{2m} - 4(1 + 2^m + 2^{2m}) \equiv 5 \cdot 2^{2m} \pmod{2^{2m+3}}.$$

This implies $v_2(t^2 - 4p) = 2m$. Applying [17, Thm. 7] as we did for odd ℓ, we see that the height of the 2-isogeny volcano corresponding to E is m, there is a unique vertex on the crater of the volcano, and this vertex corresponds to E. □

Remark 2.6. In contrast to what was just proved, If E and E' have CM then by [6, Prop. 6.1], the isogeny $E \to E'$ is either always ascending, always descending, or always horizontal, for each ordinary prime of good reduction. In Section 4 we will show that this implies that either $d_{\ell^m} = 0$ for all $m \geq 2$, or $d'_{\ell^m} = 0$ for all $m \geq 2$. We also point the interested reader to [1] for more background on the Galois theory of torsion point fields for non-CM curves

Proposition 1.2, restated here as Proposition 2.7 for convenience, now follows as an immediate corollary of Proposition 2.5 and Lemmas 2.2 and 2.4.

Proposition 2.7. *With notation as above, suppose the elliptic curves E and E' do not have CM and that the maximum of the levels of their ℓ-adic representations is ℓ^M. Then we have $d_{\ell^m}, d'_{\ell^m} \in \{0, 1 - 1/\ell\}$ for all $m \geq 1$ and $d_{\ell^m} = d'_{\ell^m} = 1 - 1/\ell$ for all $m \geq M$.*

We now prove the main result of the paper.

Theorem 2.8. *Let E and E' be ℓ-isogenous elliptic curves over $\mathbf{Q}$. Then the proportion $P(E, E')$ of primes such that $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$ is given by the formula*

$$P(E, E') = 1 - \sum_{m=1}^{\infty} \left(\frac{d_{\ell^m}}{|G(\ell^m)|} + \frac{d'_{\ell^m}}{|G'(\ell^m)|} \right). \tag{2.9}$$

Proof. The summand $d_{\ell^m}/|G(\ell^m)| + d'_{\ell^m}/|G'(\ell^m)|$ represents the proportion of primes for which one of $E(\mathbf{F}_p)$ or $E'(\mathbf{F}_p)$ has full ℓ^m-torsion, but the other does not. Summing over all m gives the proportion of primes for which $E(\mathbf{F}_p)$ and $E'(\mathbf{F}_p)$ are nonisomorphic, whence the formula (1.4). □

Theorem 2.8 holds regardless of whether E or E' has a rational point of order ℓ. If neither E nor E' has a rational point of order ℓ, then at all primes for which neither $E(\mathbf{F}_p)$ nor $E'(\mathbf{F}_p)$ has a point of order ℓ, we automatically have that $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$. We therefore expect to see high values of $P(E, E')$ in such cases. We conclude this section with an example illustrating this as well as a remark on quadratic twists.

Example 2.10. Let E be the elliptic curve 121.a1 and E' the elliptic curve 121.a2; they are 11-isogenous over $\mathbf{Q}$. Each curve has level 11 and $d_{11^m} = d'_{11^m} = 10/11$ for all $m \geq 1$. We also have $|G(11)| = |G'(11)| = 120$. Applying Theorem 2.8, we have $P(E, E') = 86509/87840 \approx 0.9848$, while $P(E, E')[10^7] \approx 0.9839$.

Checking the LMFDB, one sees that E and E' acquire an 11-torsion point over $\mathbf{Q}(\zeta_{11})^+$ and $\mathbf{Q}(\zeta_{11})$, respectively. Note that $[\mathbf{Q}(\zeta_{11}) : \mathbf{Q}] = 10$ and $[\mathbf{Q}(\zeta_{11})^+ : \mathbf{Q}] = 5$. We confirm in Pari/GP that $E(\mathbf{F}_p)$ and $E'(\mathbf{F}_p)$ have a point of order 11 for roughly $1/5$ of the primes up to 10^7 and thus for the complementary $4/5$ of the primes we automatically have $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$.

Remark 2.11. If E admits a rational ℓ-isogeny, then so does every quadratic twist E_d. However, it is not the case that $P(E, E') = P(E_d, E'_d)$, since twisting can change rational torsion. For example, if E is the elliptic curve 175.b3 and E' the curve 175.b2, then E and E' are 3-isogenous, yet neither has a rational 3-torsion point. Using the information on the 3-adic Galois images from their LMFDB pages, we compute $P(E, E') = 199/240$ via Theorem 2.8.

Twisting by 5, we check that E_5 is the curve 35.a3 and E'_5 is 35.a2, each of which has a rational 3-torsion point. Using their LMFDB data we compute $P(E_5, E'_5) = 79/120$. We also note that $199/240 > 79/120$, reflecting the fact that neither E nor E' has a rational point of order 3 and so it is more likely that $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$ than $E_5(\mathbf{F}_p) \simeq E'_5(\mathbf{F}_p)$.

3. Examples

By our discussion following the proof of Theorem 2.8, we are interested in values of $P(E, E')$ that are not artificially "inflated" by neither E nor E' having a point of order ℓ. Therefore, suppose $E(\mathbf{Q})_{\mathsf{tors}} \simeq \mathbf{Z}/\ell\mathbf{Z}$ and write $E' = E/\langle P \rangle$, where P is a rational point of order ℓ. If $\ell = 2$ then E' always has a point of order 2, while if $\ell \in \{3, 5\}$, then E' has a rational point of order ℓ only when the image of the mod ℓ representation of ℓ lies in a split Cartan subgroup of $\mathrm{GL}_2(\mathbf{Z}/\ell\mathbf{Z})$. If $\ell = 7$, then E' will never have a point of order ℓ by the main result of [8].

Given this setup, the maximal order of $G(\ell)$ and $G'(\ell)$ is $\ell(\ell - 1)$. If G and G' have level ℓ, then we have the following formula for $|G(\ell^m)|$ and $|G'(\ell^m)|$:

$$|G(\ell^m)| = |G'(\ell^m)| = (\ell - 1)\ell^{4m-3}.$$

We also have $[L_{\ell^m}L'_{\ell^m} : L_{\ell^m}] = [L_{\ell^m}L'_{\ell^m} : L'_{\ell^m}] = \ell$ so that $d_{\ell^m} = d'_{\ell^m} = 1 - 1/\ell$. Putting all of this together, when G and G' are as large as possible, we have

$$\begin{aligned} P(E,E') &= 1 - \sum_{m=1}^{\infty}\left(\frac{d_{\ell^m}}{|G(\ell^m)|} + \frac{d'_{\ell^m}}{|G'(\ell^m)|}\right) \\ &= 1 - \frac{\ell-1}{\ell}\cdot\frac{1}{\ell(\ell-1)}\sum_{m=0}^{\infty}\frac{1}{\ell^{4m}} \\ &= \frac{\ell^4 - 2\ell^2 - 1}{\ell^4 - 1}. \end{aligned}$$

Define $f(\ell) := \frac{\ell^4-2\ell^2-1}{\ell^4-1}$. Then we compute

ℓ	$f(\ell)$
2	$7/15 = 0.4\overline{6}$
3	$31/40 = 0.775$
5	$287/312 = 0.919\overline{871794}$
7	$1151/1200 = 0.9591\overline{6}$

We now give some examples illustrating Equation (1.4) when E has a rational ℓ-torsion point and G is as large as possible, as well as a special case when G is not. In all of our examples we calculate out to 10^6 and note that $\pi(10^6) = 78498$.

Example 3.1. For each of the following examples, $|G(\ell)| = |G'(\ell)| = \ell(\ell-1)$.

(1) Let E and E' be the 2-isogenous elliptic curves of the isogeny class with LMFDB label 69.a. We compute

$$P(E,E')[10^6] = \frac{36631}{\pi(10^6)} \approx 0.46665 \approx 0.4\overline{6} = f(2).$$

(2) Let E and E' be the 3-isogenous elliptic curves of the isogeny class with LMFDB label 44.a.

$$P(E,E')[10^6] = \frac{72283}{\pi(10^6)} \approx 0.77548 \approx 0.775 = f(3).$$

(3) Let E and E' be the 5-isogenous elliptic curves of the isogeny class with LMFDB label 38.b. We compute

$$P(E,E')[10^6] = \frac{60874}{\pi(10^6)} \approx 0.92082 \approx 0.919\overline{871794} = f(5).$$

(4) Let E and E' be the 7-isogenous elliptic curves of the isogeny class with LMFDB label 26.b. We compute

$$P(E,E')[10^6] = \frac{75298}{\pi(10^6)} \approx 0.95923 \approx 0.9591\overline{6} = f(7).$$

Example 3.2. Let E be the elliptic curve with LMFDB label 144.b4 and E' be 144.b3. Then $E(\mathbf{Q})_{\mathsf{tors}} \simeq E'(\mathbf{Q})_{\mathsf{tors}} \simeq \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$. Therefore $d_2 = d'_2 = 0$. According to their LMFDB pages, both elliptic curves have level 8, with $|G(4)| = 4$, $|G(8)| = 16$, $|G'(4)| = 8$, $|G'(8)| = 16$, and $L_4 = \mathbf{Q}(\zeta_{12})$. Direct calculation with Pari/GP shows that $L'_4 = \mathbf{Q}(\zeta_{24})$ and thus

$$\begin{aligned} d_4 &= 1 - X = 1 - [L_4L'_4 : L_4]^{-1} = 1/2, \text{ and} \\ d'_4 &= 1 - X' = 1 - [L_4L'_4 : L'_4]^{-1} = 0. \end{aligned}$$

Further calculation reveals that $d_{2^m} = d'_{2^m} = 1/2$ for $m = 3, 4$. By Theorem 2.8 we have

$$\begin{aligned} P(E, E') &= 1 - \sum_{m=1}^{\infty} \left(\frac{d_{2^m}}{|G(2^m)|} + \frac{d'_{2^m}}{|G'(2^m)|} \right) \\ &= 1 - \left(\frac{1/2}{|G(4)|} + \frac{1/2}{|G(8)|} \cdot \sum_{m=0}^{\infty} \frac{1}{16^m} + \frac{1/2}{|G'(8)|} \cdot \sum_{m=0}^{\infty} \frac{1}{16^m} \right) \\ &= 1 - 1/8 - \frac{1}{32} \cdot \frac{16}{15} - \frac{1}{32} \cdot \frac{16}{15} = 97/120 = 0.808\overline{3}. \end{aligned}$$

Finally, we compute

$$P(E, E')[10^6] = \frac{63469}{\pi(10^6)} \approx 0.8085.$$

4. Remarks on CM Elliptic Curves

By [11, Thm. 1], if E is an ordinary elliptic curve over a finite field $\mathbf{F}_q$ with endomorphism ring $\mathcal{O}$, then $E(\mathbf{F}_q) \simeq \mathcal{O}/(\pi - 1)$, where π represents the Frobenius as an element of $\mathcal{O}$. If E and E' are isogenous elliptic curves with endomorphism rings $\mathcal{O}$ and $\mathcal{O}'$ over $\mathbf{F}_q$, respectively, then the prime-to-$[\mathcal{O} : \mathcal{O}']$ parts of the finite groups $E(\mathbf{F}_q)$ and $E'(\mathbf{F}_q)$ are isomorphic by [3, Lem. 2]. We will study the effect of all of this in the case where E and E' define elliptic curves over $\mathbf{Q}$ with CM. Before beginning, however, we recall that by [5, Thm. 2.3.1] if E and E' have supersingular reduction then $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$. Therefore, we can focus exclusively on primes of ordinary reduction.

If E and E' have CM over $\mathbf{Q}$ then, by the Baker-Heegner-Stark Theorem, the CM orders correspond to one of the nine fundamental discriminants

$$-3, -4, -7, -8, -11, -19, -43, -67, -163,$$

or one of the four non-fundamental discriminants, $-12, -16, -27, -28$, of class number 1. Suppose that E and E' have geometric endomorphism rings $\mathcal{O}$ and $\mathcal{O}'$ over $\mathbf{Q}$, respectively. If $p > 3$ is a prime of ordinary reduction, then by [10, Ch. 13, Thm. 12] $E \pmod p$ and $E' \pmod p$ have geometric endomorphism rings $\mathcal{O}$ and $\mathcal{O}'$ as well. Because ordinary elliptic curves have all endomorphisms defined over the base field, we have that $\mathcal{O}$ and $\mathcal{O}'$ are the endomorphism rings over $\mathbf{F}_p$ of $E \pmod p$ and $E' \pmod p$, respectively.

Since E and E' are isogenous, $\mathcal{O}$ and $\mathcal{O}'$ are homothetic and, as a result, $\operatorname{disc} \mathcal{O} / \operatorname{disc} \mathcal{O}'$ is a rational square. Comparing against the possible discriminants above, those square values are either $1, 4, 9$, or their reciprocals. If $\operatorname{disc} \mathcal{O} = \operatorname{disc} \mathcal{O}'$, then $\mathcal{O} \simeq \mathcal{O}'$. This isomorphism holds modulo p as well and thus $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$.

Lemma 4.1. *Suppose E and E' are ℓ-isogenous CM elliptic curves over $\mathbf{Q}$ with $\ell > 3$. Then $P(E, E') = 1$.*

Proof. If E and E' are ℓ-isogenous, then by [9, Prop. 21] the endomorphism rings, under reduction modulo ordinary p, satisfy $[\mathcal{O} : \mathcal{O}'] = 1, \ell$, or ℓ^{-1}. However, since E and E' have CM over $\mathbf{Q}$, by our work above it follows that $[\mathcal{O} : \mathcal{O}'] \in \{1, 2^{\pm 1}, 3^{\pm 1}\}$. Therefore, we must have $\mathcal{O} \simeq \mathcal{O}'$ and hence $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$. Since $E(\mathbf{F}_p) \simeq E'(\mathbf{F}_p)$ for all supersingular primes, it follows that $P(E, E') = 1$. □

We are now reduced to studying the cases $[\mathcal{O} : \mathcal{O}'] \in \{2^{\pm 1}, 3^{\pm 1}\}$. Suppose $\ell = 2$. In [6] we proved that in the case of CM elliptic curves, the proportion of anomalous primes of a pair of 2-isogenous curves is either $1/12$ or 0. This essentially came down to the fact (proved in [6, Prop. 6.1]) that if $E \to E'$ is a 2-isogeny then either

(1) $L_{2^m} \subseteq L'_{2^m}$ for all $m \geq 2$, or
(2) $L'_{2^m} \subseteq L_{2^m}$ for all $m \geq 2$.

Thus either $d_{2^m} = 0$ for all $m \geq 2$ or $d'_{2^m} = 0$ for all $m \geq 2$. It also follows from [6, Prop. 6.1] that if $d_4 \neq 0$ (respectively $d'_4 \neq 0$), then $d_{2^m} = 1/2$ (resp. $d'_{2^m} = 1/2$) for all $m \geq 2$. The following example demonstrates how to use this property of CM elliptic curves to compute $P(E, E')$.

Example 4.2. Let E and E' be the 2-isogenous elliptic curves 49.a1 and 49.a2, respectively. These elliptic curves have CM, and one verifies that the geometric endomorphism ring of E is $\mathbf{Z}[\sqrt{-7}]$, while that of E' is $\mathbf{Z}[(-1+\sqrt{-7})/2]$.

According to their LMFDB pages, $L_2 = \mathbf{Q}(\sqrt{7})$ and $L'_2 = \mathbf{Q}(\sqrt{-7})$. Hence $d_2 = d'_2 = 1/2$. By [6, Prop. 6.1], either $d_{2^m} = 0$ for all $m \geq 2$ or $d'_{2^m} = 0$ for all $m \geq 2$. We check that $[L'_4 : L_4] = 2$, hence $d'_{2^m} = 0$ for all $m \geq 2$ and $d_{2^m} = 1/2$ for all $m \geq 2$. One checks that $|G(2^m)| = |G'(2^m)| = 2 \cdot 4^{m-1}$ for all $m \geq 2$. Thus,

$$\begin{aligned} P(E, E') &= 1 - \sum_{m=1}^{\infty} \left(\frac{d_{2^m}}{|G(2^m)|} + \frac{d'_{2^m}}{|G'(2^m)|} \right) \\ &= 1 - \left(\frac{1}{2} \sum_{m=1}^{\infty} \left(\frac{1}{2 \cdot 4^{m-1}} \right) + \frac{1}{2} \cdot \frac{1}{2} \right) \\ &= 1 - \left(\frac{1}{4} \cdot \frac{4}{3} + \frac{1}{4} \right) = \frac{5}{12} = 0.41\overline{6}. \end{aligned}$$

We separately check that $P(E, E')[10^6] \approx 0.4165$.

Similar to the non-CM case, one could work out all possible values of $P(E, E')$ for all 2-isogenous CM elliptic curves over $\mathbf{Q}$. To finish the paper, we consider the case $\ell = 3$. By the classification of isogeny-torsion graphs over $\mathbf{Q}$ in [2] and the classification of ℓ-adic Galois representations of CM elliptic curves over $\mathbf{Q}$, the isogeny-torsion graph

$$[3] \text{ —— } [3] \text{ —— } [3] \text{ —— } [0]$$

is the unique (up to isomorphism) isogeny-torsion graph over $\mathbf{Q}$ to which elliptic curves having CM discriminant -27 can belong. All such isogeny-torsion graphs contain elliptic curves that are quadratic twists of those in the isogeny-torsion graph 27.a. Moreover, by [13, Thm. 1.2, 1.4], there are only a handful of cases where E and E' are 3-isogenous, have CM, and $\operatorname{disc} \mathcal{O} / \operatorname{disc} \mathcal{O}' \in \{3^{\pm 2}\}$. We list those cases in Table 1 according to their 3-adic images G and G', discriminants, j-invariants, and rational torsion subgroups.

We conclude by working out $P(E, E')$ for an example of the third case of Table 1.

Example 4.3. Suppose E is 432.e1 and E' is 432.e4. From their LMFDB pages, one checks that $L_3 = \mathbf{Q}(\zeta_{12})$, which is an index-3 subfield of L'_3. Therefore $d_3 = 2/3$ and $d'_3 = 0$. By direct calculation, $d_9 = 2/9$ and $d'_9 = 0$. The proof of [6, Prop. 6.1] carries over immediately to the case $\ell = 3$ as well (since the endomorphism rings

$[G, \mathrm{disc}(\mathcal{O}), j, E(\mathbf{Q})_{\mathsf{tors}}]$	$[G', \mathrm{disc}(\mathcal{O}'), j', E'(\mathbf{Q})_{\mathsf{tors}}]$	Example
$[27.648.13.25, -27, -12288000, \mathbf{Z}/3\mathbf{Z}]$	$[27.1944.55.31, -3, 0, \mathbf{Z}/3\mathbf{Z}]$	27.a
$[27.648.13.34, -27, -12288000, 0]$	$[27.1944.55.37, -3, 0, \mathbf{Z}/3\mathbf{Z}]$	27.a
[maximal, −27,-12288000, 0]	$[27.972.55.16, -3, 0, 0]$	432.e

TABLE 1.

of $E \pmod p$ and $E' \pmod p$ are fixed for all ordinary p, the rational 3-isogeny $E \to E'$ always reduces to an ascending 3-isogeny), hence $d'_{3^m} = 0$ for all m and $d_{3^m} = 2/3$ for all m. We have $|G(3^m)| = 4 \cdot 9^{m-1}$ for all $m \geq 1$. Now we compute:

$$P(E, E') = 1 - \sum_{m=1}^{\infty} \left(\frac{d_{3^m}}{|G(3^m)|} + \frac{d'_{3^m}}{|G'(3^m)|} \right) = 1 - \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{10}{9} = \frac{22}{27} = 0.\overline{814}.$$

For comparison, a routine calculation shows that $P(E, E')[10^7] \approx 0.8126$.

ACKNOWLEDGMENTS

The second author was supported by DMS 2154223.

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